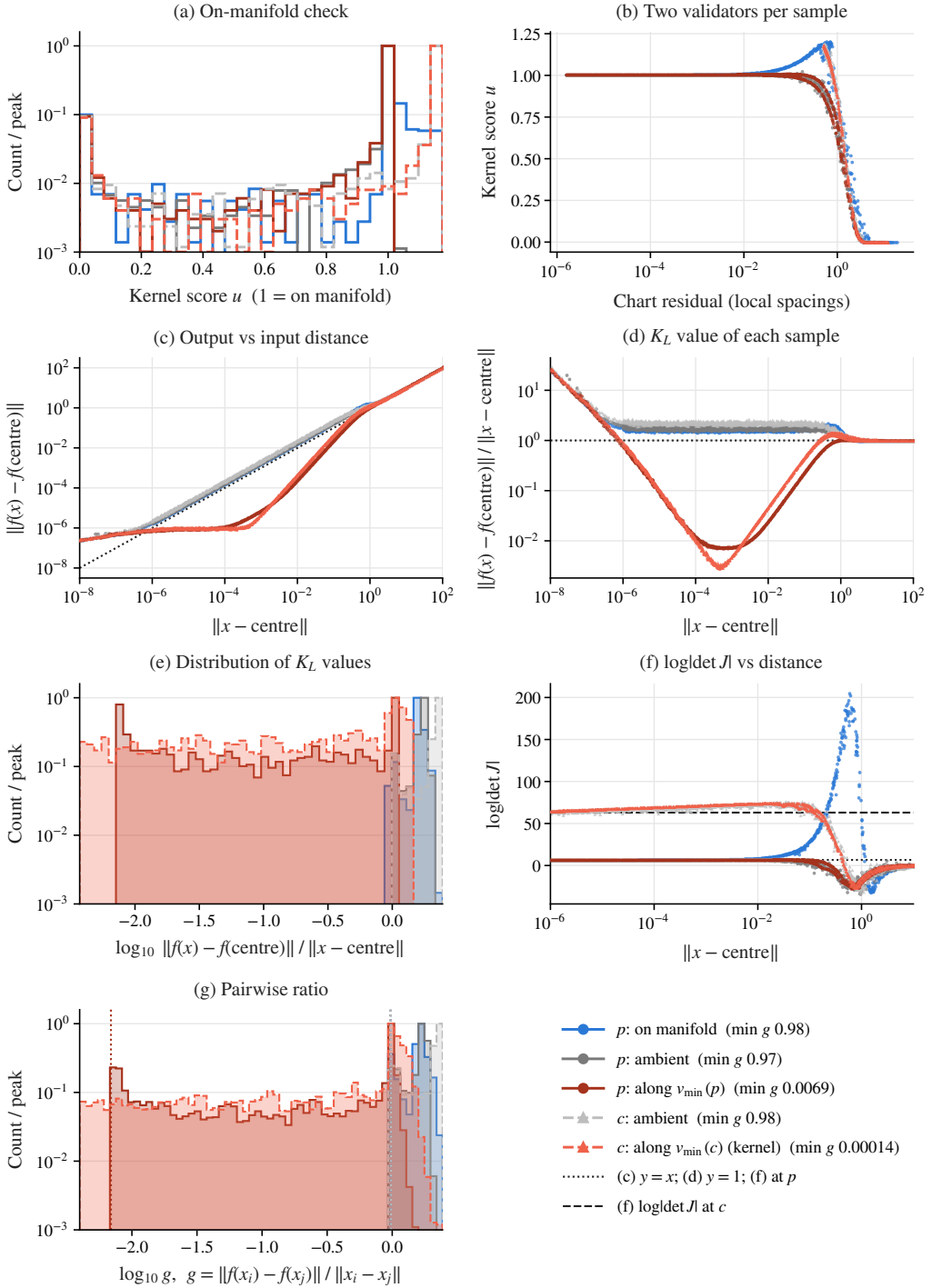

(a) On-manifold check
Count / peak
Kernel score u (1 = on manifold)
(b) Two validators per sample
Kernel score u
Chart residual (local spacings)
(c) Output vs input distance
$\|f(x) - f(\text{centre})\|$
$\|x - \text{centre}\|$
(d) K_L value of each sample
$\|f(x) - f(\text{centre})\| / \|x - \text{centre}\|$
$\|x - \text{centre}\|$
(e) Distribution of K_L values
Count / peak
$\log_{10} \|f(x) - f(\text{centre})\| / \|x - \text{centre}\|$
(f) log|det J| vs distance
log|det J|
$\|x - \text{centre}\|$
(g) Pairwise ratio
Count / peak
$\log_{10} g$, $g = \|f(x_i) - f(x_j)\| / \|x_i - x_j\|$
p: on manifold (min g 0.98)
p: ambient (min g 0.97)
p: along $v_{\min}(p)$ (min g 0.0069)
c: ambient (min g 0.98)
c: along $v_{\min}(c)$ (kernel) (min g 0.00014)
(c) $y = x$; (d) $y = 1$; (f) at p
(f) log|det J| at c